U(3) Structure on the sixPlaneAlpha Manifold

Defining the U(3) Structure on the 6-Dimensional Intersection Plane arising from the Pre-Universe Cl(4,4) Spinor Geometry

Patrick L. Nash (c) 2022-2025, GPL; based on Dirac-Lord-Nash_4+4.wl

In[1]:=
```
timeStart = Now
```

Out[1]= Mon 30 Mar 2026 06:25:56 GMT−7

In[2]:=
```
Print[timeStart];
```

Mon 30 Mar 2026 06:25:56 GMT−7

Section 0: Prerequisites

Load the Clifford algebra Cl(4,4) package which defines: η88 (8x8 split metric), γ88[A] (8x8 generators), CA8[n] (16x16 generators), σ88/sigma16 (chirality matrices), Ω8/Ω16 (complex structure matrices), SAB16[A,B] (Spin generators), unit (unit type-1 spinor), FAa/FaA (F-matrices), and the 256-element Pauli basis.

In[3]:=
```
ClearAll["Global`*"];
Get[FileNameJoin[{NotebookDirectory[], "Dirac-Lord-Nash_4+4-REVA.wl"}]];
```

```
Print[basisVectorSplitOctonion] = {𝕖0, 𝕖1, 𝕖2, 𝕖3, 𝕖4, 𝕖5, 𝕖6, 𝕖7}
```

```
Dirac-Lord-Nash_4+4 loaded successfully!
  BUT, WARNING:  DO NOT USE IF YOU WANT A CORRECT RESULT!
```

In[5]:=
```
(*CHECK*)
Ω8 === σ88 . γ88[7]
η88 === DiagonalMatrix[{1, 1, 1, 1, -1, -1, -1, -1}]
```

Out[5]= True

Out[6]= True

In[7]:=
```
verifyAntiCommutation8[]
```

Out[7]= True

In[8]:=
```
verifyAntiCommutation16[]
```

Out[8]= True

In[9]:=
```
verifySABCA8Commutation[]
```
Out[9]=
```
{True}
```

In[10]:=
```
verifySO44Commutation[]
```
Out[10]=
```
{True}
```

---

some function definitions :

In[11]:=
```
expr_* := expr /. Complex[a_, b_] :> Complex[a, -b]
```

In[12]:=
```
ra[z_] := 1/2 (z + (z /. {Complex[a_, b_] :> Complex[a, -b]}))
```

In[13]:=
```
ia[z_] := 1/(2 i) (z - (z /. {Complex[a_, b_] :> Complex[a, -b]}))
```

GENERALIZED KroneckerDelta :

In[14]:=
```
Clear[kδ];
kδ[lower_, upper_] /; Length[lower] == Length[upper] :=
 Det[Outer[KroneckerDelta, lower, upper]]
```

In[15]:=
```
{kδ[{1}, {2}], kδ[{1}, {1}], kδ[{1, 2}, {2, 1}]}
```
Out[15]=
```
{0, 1, -1}
```

In[16]:=
```
kδ[{1}, {1}]
```
Out[16]=
```
1
```

In[17]:=
```
Clear[rawSets];
rawSets[l_, f_, o_ : 0] := Module[{t, h},
  t = Table[Cases[
 l[[j]],
h_Symbol[n_Integer] /; StringEndsQ[SymbolName[h], f] :> n + o,
  Infinity,
  Heads → True
], {j, 1, Length[l]}];
  Return[t];
 ]
```

In[19]:=
```
Clear[dualRawSets];
dualRawSets[l_, f_, o_, h_, oh_] := Module[{r, r1, r2},
  r1 = rawSets[l, f, o];
  r2 = rawSets[l, h, oh];
  r = Union[Flatten[{r1, r2}, 1]] // Drop[#, 1] &;
  Return[r];
 ]
```

'FixedPoint', below, uses 'MergeSetsStep', below.
According to Wolfram docs, one uses 'Throw' to exit from 'FixedPoint' before it is finished.
FixedPoint really uses Catch and Throw, so what could go wrong

Details and Options

- FixedPoint always returns the last result it gets.
- You can use Throw to exit from FixedPoint before it is finished.
- FixedPoint[*f*, *expr*] applies SameQ to successive pairs of results to determine whether a fixed point has been reached.
- FixedPoint[*f*, *expr*, …, SameTest –> *s*] applies *s* to successive pairs of results.

In[21]:=

```
(* Helper to perform one pass of merging *)
```

Delete[*expr*, {*i*, *j*, …}] deletes the part at position {*i*, *j*, …}.

```
MergeSetsStep[currentSets_List] :=
  Module[{i, j, n = Length[currentSets], merged = currentSets},
 (* We look for the first pair (i, j) that intersects *)
 Catch[
  For[i = 1, i ≤ Length[merged], i++,
   For[j = i + 1, j ≤ Length[merged], j++,
    If[Intersection[merged⟦i⟧, merged⟦j⟧] =!= {},

          (* Found intersection: Merge and Throw to restart/finish this step *)
     merged = Delete[merged, {{i}, {j}}];
     AppendTo[merged, Sort[Union[currentSets⟦i⟧, currentSets⟦j⟧]]];
     Throw[Sort[merged]]; (* Sort for canonical form *)
    ];
   ];
  ];
  (* If no intersection found, return original sorted *)
  Sort[merged]
 ]
];
```

Out[21]=

RefLink[Delete, paclet: ref/Delete][expr, {i, j, …}] deletes the part at position {i, j, …}.

In[23]:=

```
showCoupledEquations[items_List] := FixedPoint[MergeSetsStep, items];
```

In[24]:=

```
Clear[safeInverse];
safeInverse[m_?MatrixQ] := Module[{a, b},
  {a, b} = Dimensions[m];
  Catch[If[a ≠ b, Throw[Return[{a, b}]], ## &[]]];
  Catch[If[Det[m] == 0, Throw[Return["Error: Singular Matrix"]];, ## &[]]];
  Return[ Inverse[m]];];
```

In[26]:=
```
blockPartition[mat_] :=
 Module[{(*blocks,block11,block12,block21,block22,*)ret},
  blocks = Partition[mat, {8, 8}];
  (*block11=blocks⟦1,1⟧ ;*)(*Top-left block*)
  (*block12=blocks⟦1,2⟧ ;*)(*Top-right block*)
  (*block21=blocks⟦2,1⟧ ;*)(*Bottom-left block*)
  (*block22=blocks⟦2,2⟧ ;*)(*Bottom-right block*)
  ret = ArrayFlatten[
     {{blocks⟦1, 1⟧, blocks⟦1, 2⟧}, {blocks⟦2, 1⟧, blocks⟦2, 2⟧}}] === mat;
  If[ret, Return[
    {{blocks⟦1, 1⟧, blocks⟦1, 2⟧}, {blocks⟦2, 1⟧, blocks⟦2, 2⟧}}], ## &[]];
  Return[Null];]
```

In[27]:=
```
eextract[a_] := Extract[a, {#}] & /@ Range[0, Length[a]]
```

Sequence[Subscript[expr, 1], Subscript[expr, 2], …]
represents a sequence of arguments to be spliced automatically into any function .

In[28]:=
```
skelx[x_, matQ_ : True] := Module[{t = {666}},
  If[matQ == True, t = Block[{r},
     r = x;
     Table[Block[{q}, q = SameQ[0, r⟦j, i⟧];
       If[True == q, Style[0, Blue], Style[Length[r⟦j, i⟧], Red]]],
      {j, 1, Length[r]}, {i, 1, Length[r⟦-1⟧]}]]];
  If[matQ == False, t = Block[{r},
     r = x;
     Table[Block[{q}, q = SameQ[0, r⟦j⟧];
       If[True == q, Style[0, {RGBColor → {0, 1 / 3, 0}, Bold}],
        Style[Length[r⟦j⟧], Red]]], {j, 1, Length[r]}]]];
  MatrixForm[t, TableAlignments → Left]]
```

In[29]:=
```
makeSym[size_, fn_] :=
 Module[{rtmp}, rtmp = Table[fn[i, j], {i, 1, size}, {j, 1, i}];
  MapThread[Join, {rtmp, Rest /@ Flatten[rtmp, {{2}, {1}}]}]]
```

In[30]:=
```
makeAntiSym[size_, fn_] :=
 Module[{rtmp}, rtmp = Table[If[i == j, 0, fn[i, j]], {i, 1, size}, {j, 1, i}];
  MapThread[Join, {rtmp, Rest /@ Flatten[-rtmp, {{2}, {1}}]}]]
```

In[31]:=
```
Block[{MX}, MX = makeAntiSym[8, Subscript[ωω, ##] &]]
```

Out[31]=

$$\{\{0, -\omega\omega_{2,1}, -\omega\omega_{3,1}, -\omega\omega_{4,1}, -\omega\omega_{5,1}, -\omega\omega_{6,1}, -\omega\omega_{7,1}, -\omega\omega_{8,1}\},$$
$$\{\omega\omega_{2,1}, 0, -\omega\omega_{3,2}, -\omega\omega_{4,2}, -\omega\omega_{5,2}, -\omega\omega_{6,2}, -\omega\omega_{7,2}, -\omega\omega_{8,2}\},$$
$$\{\omega\omega_{3,1}, \omega\omega_{3,2}, 0, -\omega\omega_{4,3}, -\omega\omega_{5,3}, -\omega\omega_{6,3}, -\omega\omega_{7,3}, -\omega\omega_{8,3}\},$$
$$\{\omega\omega_{4,1}, \omega\omega_{4,2}, \omega\omega_{4,3}, 0, -\omega\omega_{5,4}, -\omega\omega_{6,4}, -\omega\omega_{7,4}, -\omega\omega_{8,4}\},$$
$$\{\omega\omega_{5,1}, \omega\omega_{5,2}, \omega\omega_{5,3}, \omega\omega_{5,4}, 0, -\omega\omega_{6,5}, -\omega\omega_{7,5}, -\omega\omega_{8,5}\},$$
$$\{\omega\omega_{6,1}, \omega\omega_{6,2}, \omega\omega_{6,3}, \omega\omega_{6,4}, \omega\omega_{6,5}, 0, -\omega\omega_{7,6}, -\omega\omega_{8,6}\},$$
$$\{\omega\omega_{7,1}, \omega\omega_{7,2}, \omega\omega_{7,3}, \omega\omega_{7,4}, \omega\omega_{7,5}, \omega\omega_{7,6}, 0, -\omega\omega_{8,7}\},$$
$$\{\omega\omega_{8,1}, \omega\omega_{8,2}, \omega\omega_{8,3}, \omega\omega_{8,4}, \omega\omega_{8,5}, \omega\omega_{8,6}, \omega\omega_{8,7}, 0\}\}$$

---

split octonions; evalues, evecs of $\sigma 88$; split octonion multiplication constants

$$m^{c}_{ab} = \mathbb{F}^{A}_{a}\, \tau_{A}{}^{c}{}_{b}$$
$$m^{C}_{AB} = \mathbb{F}^{C}_{c}\, \tau_{A}{}^{c}{}_{b}\, \mathbb{F}^{b}_{B}$$

In[32]:=
```
splitOctonionMultTableRETRO
```

Out[32]=

times	eA[1]	eA[2]	eA[3]	eA[4]	eA[5]	eA[6]	eA[7]	eA[8]
eA[1]	eA[1]	eA[2]	eA[3]	eA[4]	eA[5]	eA[6]	eA[7]	eA[8]
eA[2]	eA[2]	-eA[1]	-eA[4]	eA[3]	-eA[6]	eA[5]	eA[8]	-eA[7]
eA[3]	eA[3]	eA[4]	-eA[1]	-eA[2]	-eA[7]	-eA[8]	eA[5]	eA[6]
eA[4]	eA[4]	-eA[3]	eA[2]	-eA[1]	-eA[8]	eA[7]	-eA[6]	eA[5]
eA[5]	eA[5]	eA[6]	eA[7]	eA[8]	eA[1]	eA[2]	eA[3]	eA[4]
eA[6]	eA[6]	-eA[5]	eA[8]	-eA[7]	-eA[2]	eA[1]	-eA[4]	eA[3]
eA[7]	eA[7]	-eA[8]	-eA[5]	eA[6]	-eA[3]	eA[4]	eA[1]	-eA[2]
eA[8]	eA[8]	eA[7]	-eA[6]	-eA[5]	-eA[4]	-eA[3]	eA[2]	eA[1]

In[33]:=
```
(*`Public`Subscript[`Public`e, 0];*)
```

In[34]:=
```
ToExpression["Subscript[e," <> ToString[#] <> "]"] & /@ Range[0, 7]
```

Out[34]=

$$\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

In[35]:=
```
(*basisVectorSplitOctonion=
 ToExpression["Subscript[e,"<>ToString[#]<>"]"]&/@Range[0,7]*)
```

In[36]:=
```
splitOctonionMultTableB[
 ToExpression["Subscript[e," <> ToString[#] <> "]"] & /@ Range[0, 7]]
```

Out[36]=

A/B	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	$-e_3$	e_2	$-e_5$	e_4	e_7	$-e_6$
e_2	e_2	e_3	$-e_0$	$-e_1$	$-e_6$	$-e_7$	e_4	e_5
e_3	e_3	$-e_2$	e_1	$-e_0$	$-e_7$	e_6	$-e_5$	e_4
e_4	e_4	e_5	e_6	e_7	e_0	e_1	e_2	e_3
e_5	e_5	$-e_4$	e_7	$-e_6$	$-e_1$	e_0	$-e_3$	e_2
e_6	e_6	$-e_7$	$-e_4$	e_5	$-e_2$	e_3	e_0	$-e_1$
e_7	e_7	e_6	$-e_5$	$-e_4$	$-e_3$	$-e_2$	e_1	e_0

In[37]:=
```
{evalues, evecs} = Eigensystem[σ88]
```

Out[37]=
```
{{-1, -1, -1, -1, 1, 1, 1, 1}, {{0, 0, 0, -1, 0, 0, 0, 1}, {0, 0, -1, 0, 0, 0, 1, 0},
  {0, -1, 0, 0, 0, 1, 0, 0}, {-1, 0, 0, 0, 1, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 1},
  {0, 0, 1, 0, 0, 0, 1, 0}, {0, 1, 0, 0, 0, 1, 0, 0}, {1, 0, 0, 0, 1, 0, 0, 0}}}
```

In[38]:=
```
(u = ExpandAll[1/Sqrt[2] evecs]) // MatrixForm
```

Out[38]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 \end{pmatrix}$$

In[39]:=
```
Table[u[[h]].σ88.u[[h]], {h, 1, Length[u]}]
```

Out[39]=
```
{-1, -1, -1, -1, 1, 1, 1, 1}
```

In[40]:=
```
Table[(Transpose[u][[h]]).σ88.u[[h]], {h, 1, Length[u]}]
```

Out[40]=
```
{-1, -1, -1, -1, 1, 1, 1, 1}
```

In[41]:=
```
Table[(Transpose[𝔲][[A]]) === -σ88.𝔲[[A]], {A, 1, 4}]
Table[(Transpose[𝔲][[A]]) === σ88.𝔲[[A]], {A, 5, Length[𝔲]}]
```

Out[41]=
```
{True, True, True, True}
```

Out[42]=
```
{True, True, True, True}
```

In[43]:=
```
hUSE = 8;
```

In[44]:=
```
Protect[hUSE]
```

Out[44]=
```
{hUSE}
```

In[45]:=
```
unit === 𝔲[[hUSE]]
```

Out[45]=
```
True
```

In[46]:=
```
(*Protect[unit]*)
```

In[47]:=
```
(Transpose[unit].σ88.unit)
```

Out[47]=
```
1
```

In[48]:=
```
(Transpose[unit].σ88.(γ̄88[0]))
```

Out[48]=

$$\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0\right\}$$

In[49]:=
```
FAa
```

Out[49]=

$$\left\{\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}\right\}, \left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, \frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}\right\}, \left\{\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0\right\}\right\}$$

In[50]:=
```
FaA
```

Out[50]=

$$\left\{\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, \frac{1}{\sqrt{2}}\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0, 0\right\}, \left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0\right\}, \left\{\frac{1}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, -\frac{1}{\sqrt{2}}\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, 0, 0\right\}, \left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\}\right\}$$

In[51]:=
```
FaA.FAa === FAa.FaA === ID8(*//Simplify//MatrixForm*)
```

Out[51]=

True

In[52]:=
```
Do[E^A_a[h] = Table[(𝕦[[h]].σ88.(γ̄88[B])), {B, 0, 7}], {h, 1, Length[𝕦]}];
```

$\mathfrak{F}^A_a = \frac{1}{\sqrt{\tilde{u}\sigma u}} u^c \sigma_{cb} \bar{\tau}^{Ab}{}_a$

```
Do[E^a_A[h] = FullSimplify[Inverse[E^A_a[h]]], {h, 1, Length[𝕦]}];
```

$\mathfrak{F}^a_A = \frac{1}{\sqrt{\tilde{u}\sigma u}} \tau_A{}^a{}_b u^b$

```
Table[E^a_A[h] === (𝕦[[h]].σ88.𝕦[[h]])
   Transpose[Table[FullSimplify[ExpandAll[η88[[B, B]] × γ88[B - 1].𝕦[[h]] ]],
      {B, 1, 8}]], {h, 1, Length[𝕦]}]
```

Out[52]=

$$\mathfrak{F}^A_a = \frac{1}{\sqrt{\tilde{u}\sigma u}} u^c \sigma_{cb} \bar{\tau}^{Ab}{}_a$$

Out[53]=

$$\mathfrak{F}^a_A = \frac{1}{\sqrt{\tilde{u}\sigma u}} \tau_A{}^a{}_b u^b$$

Out[54]=

{True, True, True, True, True, True, True, True}

Lemma [19]. Let T and W be any 8×8 matrices satisfying

(2.14) $$(\sigma T)^{\mathrm{T}} = \sigma T\,,$$

where $(\ )^{\mathrm{T}}$ denotes the transpose and

(2.15) $$(W\sigma^{-1})^{\mathrm{T}} = W\sigma^{-1}\,,$$

and moreover transforming under $\overline{SO(4,4)}$ according to

(2.16) $$T \to D^{(1)}\,T D^{(1)-1}$$

and

(2.17) $$W \to D^{(2)-1}\,W D^{(2)}\,.$$

Then

(2.18) $$\tau_{A'}\,T\,\bar{\tau}^{A'} = I\,\mathrm{tr}\,(T)$$

and

(2.19) $$\bar{\tau}^{A'}\,\widetilde{W}\,\tau_{A'} = I\,\mathrm{tr}\,(W)\,.$$

In[55]:=
```
γ88[0]
```

Out[55]=
```
{{1, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0},
 {0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 0},
 {0, 0, 0, 0, 0, 1, 0, 0}, {0, 0, 0, 0, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 0, 0, 1}}
```

In[56]:=
```
nashID[MX_] :=
 Tr[MX] * ID8 ===  Sum[γ̄88[A - 1].MX.γ88[A - 1] η88[[A, A]], {A, 1, 8}]
nashIDT[MX_] :=
 Tr[MX] * ID8 === Sum[γ88[A - 1].MX.γ̄88[A - 1] × η88[[A, A]], {A, 1, 8}]
```

In[58]:=
```
Dimensions[SAB]
```

Out[58]=
```
{8, 8, 16, 16}
```

In[59]:=
```
Block[{A1 = 1, B1 = 2}, σ1616.SAB⟦A1, B1⟧ // MatrixForm]
```

Out[59]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0
\end{pmatrix}$$

In[60]:=
```
Table[{{A1, B1}, σ88.Ω8.SAB1⟦A1, B1⟧ === Transpose[σ88.Ω8.SAB1⟦A1, B1⟧]},
 {A1, 1, 7}, {B1, A1 + 1, 8}]
```

Out[60]=
```
{{{{1, 2}, True}, {{1, 3}, True}, {{1, 4}, True}, {{1, 5}, False}, {{1, 6}, False},
  {{1, 7}, False}, {{1, 8}, True}}, {{{2, 3}, False}, {{2, 4}, False},
  {{2, 5}, True}, {{2, 6}, True}, {{2, 7}, True}, {{2, 8}, False}},
 {{{3, 4}, False}, {{3, 5}, True}, {{3, 6}, True}, {{3, 7}, True}, {{3, 8}, False}},
 {{{4, 5}, True}, {{4, 6}, True}, {{4, 7}, True}, {{4, 8}, False}},
 {{{5, 6}, False}, {{5, 7}, False}, {{5, 8}, True}},
 {{{6, 7}, False}, {{6, 8}, True}}, {{{7, 8}, True}}}
```

In[61]:=
```
nashIDT[Ω8.SAB1⟦#⟦1⟧, #⟦2⟧⟧] & /@
 {{1, 2}, {1, 3}, {1, 4}, {4, 5}, {4, 6}, {4, 7}, {5, 8}, {6, 8}, {7, 8}}
```

Out[61]=
```
{True, True, True, True, True, True, True, True, True}
```

In[62]:=
```
Table[{{A1, B1}, σ1616.CA8[8].SAB⟦A1, B1⟧ === -Transpose[σ1616.CA8[8].SAB⟦A1, B1⟧]},
 {A1, 1, 7}, {B1, A1 + 1, 8}]
```

Out[62]=
```
{{{{1, 2}, True}, {{1, 3}, True}, {{1, 4}, True}, {{1, 5}, True},
  {{1, 6}, True}, {{1, 7}, True}, {{1, 8}, True}}, {{{2, 3}, True},
  {{2, 4}, True}, {{2, 5}, True}, {{2, 6}, True}, {{2, 7}, True}, {{2, 8}, True}},
 {{{3, 4}, True}, {{3, 5}, True}, {{3, 6}, True}, {{3, 7}, True}, {{3, 8}, True}},
 {{{4, 5}, True}, {{4, 6}, True}, {{4, 7}, True}, {{4, 8}, True}},
 {{{5, 6}, True}, {{5, 7}, True}, {{5, 8}, True}},
 {{{6, 7}, True}, {{6, 8}, True}}, {{{7, 8}, True}}}
```

In[63]:=
```
Table[{{A1, B1}, σ1616.SAB⟦A1, B1⟧ === -Transpose[σ1616.SAB⟦A1, B1⟧]},
  {A1, 1, 7}, {B1, A1 + 1, 8}] // MatrixForm
```

Out[63]//MatrixForm=
```
( {{{1, 2}, True}, {{1, 3}, True}, {{1, 4}, True}, {{1, 5}, True}, {{1, 6}, True}, {{1, 7},
        {{{2, 3}, True}, {{2, 4}, True}, {{2, 5}, True}, {{2, 6}, True}, {{2, 7}, True},
                {{{3, 4}, True}, {{3, 5}, True}, {{3, 6}, True}, {{3, 7}, True}, {{3, 8},
                        {{{4, 5}, True}, {{4, 6}, True}, {{4, 7}, True}, {{4, 8}, True}}
                                {{{5, 6}, True}, {{5, 7}, True}, {{5, 8}, True}}
                                        {{{6, 7}, True}, {{6, 8}, True}}
                                                {{{7, 8}, True}}
```

```
Union[Table[Sum[γ88[A].bas64⟦k, 1⟧.σ88.γ̄88[A] × η8⟦A, A⟧, {A, 1, 8}] -
   Id Tr[bas64⟦k, 1⟧.σ88 ] +
   4 eg⟦k⟧ (bas64⟦k, 1⟧ - Transpose[bas64⟦k, 1⟧])/2 .σ88, {k, 1, 64}]]
```

In[64]:=
```
γ̄88[0]
```

Out[64]=
```
{{1, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0},
 {0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 0},
 {0, 0, 0, 0, 0, 1, 0, 0}, {0, 0, 0, 0, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 0, 0, 1}}
```

In[65]:=
```
(Transpose[unit].σ88.(γ̄88[0]))
```

Out[65]=
$$\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0\right\}$$

In[ ]:=
```
F^a_A = Transpose[η88⟦# + 1, # + 1⟧ * (γ88[#].unit) & /@ Range[0, 7]]
```

During evaluation of In[ ]:=

Set: "Tag \!\(*RowBox[{\"Power\"}]\) in \!\(*SubsuperscriptBox[\"F\", \"A\", \"a\"]\) is Protected."

Out[ ]=
$$\left\{\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, \frac{1}{\sqrt{2}}\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0, 0\right\},\right.$$

$$\left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0\right\},$$
$$\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, -\frac{1}{\sqrt{2}}\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, 0, 0\right\},$$
$$\left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\}\}$$

In[ ]:=
```
F^A_a = Transpose[unit].σ88.(γ̄88[#]) & /@ Range[0, 7]
```

During evaluation of In[ ]:=

Set: "Tag \!\(*RowBox[{\"Power\"}]\) in \!\(*SubsuperscriptBox[\"F\", \"a\", \"A\"]\) is Protected."

Out[ ]=

$$\left\{\left\{\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0\right\}, \left\{0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}\right\},\right.$$
$$\left\{0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\}, \left\{0, -\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\},$$
$$\left\{0, \frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0\right\}, \left\{0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0\right\},$$
$$\left.\left\{0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}\right\}, \left\{\frac{1}{\sqrt{2}}, 0, 0, 0, -\frac{1}{\sqrt{2}}, 0, 0, 0\right\}\right\}$$

---

defs; some Symbols; metric

Introduce the wave function, Ψ16, for a spacetime:

In[66]:=
```
X = ToExpression["x" <> ToString[#]] & /@ Range[0, 7]
```

Out[66]=
```
{x0, x1, x2, x3, x4, x5, x6, x7}
```

In[67]:=
```
K = ToExpression["k" <> ToString[#]] & /@ Range[0, 7]
```

Out[67]=
```
{k0, k1, k2, k3, k4, k5, k6, k7}
```

In[68]:=
```
Protect[X, K, x0, x1, x2, x3, x4, x5, x6, x7, k0, k1, k2, k3, k4, k5, k6, k7]
```

Out[68]=
```
{X, K, x0, x1, x2, x3, x4, x5, x6, x7, k0, k1, k2, k3, k4, k5, k6, k7}
```

In[69]:=
```
K.X
sX = Thread[X → Table[ToExpression["#" <> ToString[j]], {j, 0, 7}]]
%% /. %
sKX = Plus @@ %
sEKX = Exp[ⅈ %]
```

Out[69]=
```
k0 x0 + k1 x1 + k2 x2 + k3 x3 + k4 x4 + k5 x5 + k6 x6 + k7 x7
```

Out[70]=
```
{x0 → #0, x1 → #1, x2 → #2, x3 → #3, x4 → #4, x5 → #5, x6 → #6, x7 → #7}
```

Out[71]=
```
k0 #0 + k1 #1 + k2 #2 + k3 #3 + k4 #4 + k5 #5 + k6 #6 + k7 #7
```

Out[72]=
```
k0 #0 + k1 #1 + k2 #2 + k3 #3 + k4 #4 + k5 #5 + k6 #6 + k7 #7
```

Out[73]=

$e^{i\,(k0\,\#0+k1\,\#1+k2\,\#2+k3\,\#3+k4\,\#4+k5\,\#5+k6\,\#6+k7\,\#7)}$

In[74]:=
```
(Ψ16 = Table[f16[j][##] & @@ X, {j, 0, 15}]) // Column
```

Out[74]=
```
f16[0][x0, x1, x2, x3, x4, x5, x6, x7]
f16[1][x0, x1, x2, x3, x4, x5, x6, x7]
f16[2][x0, x1, x2, x3, x4, x5, x6, x7]
f16[3][x0, x1, x2, x3, x4, x5, x6, x7]
f16[4][x0, x1, x2, x3, x4, x5, x6, x7]
f16[5][x0, x1, x2, x3, x4, x5, x6, x7]
f16[6][x0, x1, x2, x3, x4, x5, x6, x7]
f16[7][x0, x1, x2, x3, x4, x5, x6, x7]
f16[8][x0, x1, x2, x3, x4, x5, x6, x7]
f16[9][x0, x1, x2, x3, x4, x5, x6, x7]
f16[10][x0, x1, x2, x3, x4, x5, x6, x7]
f16[11][x0, x1, x2, x3, x4, x5, x6, x7]
f16[12][x0, x1, x2, x3, x4, x5, x6, x7]
f16[13][x0, x1, x2, x3, x4, x5, x6, x7]
f16[14][x0, x1, x2, x3, x4, x5, x6, x7]
f16[15][x0, x1, x2, x3, x4, x5, x6, x7]
```

In[75]:=
```
(*Table[ToExpression["((Q["<>ToString[j]<>"])&)"],{j,0,15}]*)
```

In[76]:=
```
(*Clear[sf16Aa];
sfψ16Aa=f16[#]→ToExpression["((Z["<>ToString[#]<>"][6*H*#1,H*#2])&)"]&/@
  Range[0,15]*)
```

---

vectorFormOfType1 is dual to 7-plane 7_1 ;
vectorFormOfType2 is dual to 7-plane 7_2 ;
Assuming that 7_1 and 7_2 are not parallel,
M6 = $7_1 \cap 7_2$
M6 has 28 parameterizations;

ℝ = linear span of eA[1];

Define $\psi_1, \psi_2 \in \mathbb{O}_{4,4}$

In[77]:=
```
ψ1 = Ψ16[[1 ;; 8]]
```
Out[77]=
```
{f16[0][x0, x1, x2, x3, x4, x5, x6, x7], f16[1][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[2][x0, x1, x2, x3, x4, x5, x6, x7], f16[3][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[4][x0, x1, x2, x3, x4, x5, x6, x7], f16[5][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[6][x0, x1, x2, x3, x4, x5, x6, x7], f16[7][x0, x1, x2, x3, x4, x5, x6, x7]}
```

In[78]:=
```
ψ2 = Ψ16[[8 + 1 ;; 8 + 8]]
```
Out[78]=
```
{f16[8][x0, x1, x2, x3, x4, x5, x6, x7], f16[9][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[10][x0, x1, x2, x3, x4, x5, x6, x7], f16[11][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[12][x0, x1, x2, x3, x4, x5, x6, x7], f16[13][x0, x1, x2, x3, x4, x5, x6, x7],
 f16[14][x0, x1, x2, x3, x4, x5, x6, x7], f16[15][x0, x1, x2, x3, x4, x5, x6, x7]}
```

In[79]:=
```
realPartOfType1 = unit.ψ1 // FullSimplify
```
Out[79]=

$$\frac{\text{f16[0][x0, x1, x2, x3, x4, x5, x6, x7]} + \text{f16[4][x0, x1, x2, x3, x4, x5, x6, x7]}}{\sqrt{2}}$$

In[80]:=
```
realPartOfType2 = unit.ψ2 // FullSimplify
```
Out[80]=

$$\frac{\text{f16[8][x0, x1, x2, x3, x4, x5, x6, x7]} + \text{f16[12][x0, x1, x2, x3, x4, x5, x6, x7]}}{\sqrt{2}}$$

two 7 - planes , each dual to vectorFormOfType2, vectorFormOfType1, respectively;
intersection of planes , if not parallel, yields a 6-plane M^6

remember, one split octonion algebra $\mathbb{O}_{4,4}$; $\psi_1, \psi_2 \in \mathbb{O}_{4,4}$

vector form of type 1 and 2 spinors

$$\mathfrak{F}_A^a = \frac{1}{\sqrt{\tilde{u}\sigma u}} \tau_A{}^a{}_b u^b \tag{63}$$

$$\mathfrak{F}_a^A = \frac{1}{\sqrt{\tilde{u}\sigma u}} u^c \sigma_{cb} \bar{\tau}^{Ab}{}_a \tag{64}$$

In[81]:=
```
FAa.FaA === FaA.FAa === ID8
```
Out[81]=
```
True
```

In[82]:=
```
(vectorFormOfType1 = (η88.(Transpose[ψ1].σ88.FaA)) // FullSimplify) // Column
```
Out[82]=

$$\frac{f16[0][x0,x1,x2,x3,x4,x5,x6,x7]+f16[4][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$-\frac{f16[3][x0,x1,x2,x3,x4,x5,x6,x7]+f16[7][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[2][x0,x1,x2,x3,x4,x5,x6,x7]+f16[6][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$-\frac{f16[1][x0,x1,x2,x3,x4,x5,x6,x7]+f16[5][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[1][x0,x1,x2,x3,x4,x5,x6,x7]-f16[5][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{-f16[2][x0,x1,x2,x3,x4,x5,x6,x7]+f16[6][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{-f16[3][x0,x1,x2,x3,x4,x5,x6,x7]+f16[7][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[0][x0,x1,x2,x3,x4,x5,x6,x7]-f16[4][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$

In[83]:=
```
(vectorFormOfType2 = FAa.ψ2 // FullSimplify) // Column
```
Out[83]=

$$\frac{f16[8][x0,x1,x2,x3,x4,x5,x6,x7]+f16[12][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$-\frac{f16[11][x0,x1,x2,x3,x4,x5,x6,x7]+f16[15][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[10][x0,x1,x2,x3,x4,x5,x6,x7]+f16[14][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$-\frac{f16[9][x0,x1,x2,x3,x4,x5,x6,x7]+f16[13][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[9][x0,x1,x2,x3,x4,x5,x6,x7]-f16[13][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{-f16[10][x0,x1,x2,x3,x4,x5,x6,x7]+f16[14][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{-f16[11][x0,x1,x2,x3,x4,x5,x6,x7]+f16[15][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$
$$\frac{f16[8][x0,x1,x2,x3,x4,x5,x6,x7]-f16[12][x0,x1,x2,x3,x4,x5,x6,x7]}{\sqrt{2}}$$

---

7 - planes, dual to vectors vectorFormOfType2, vectorFormOfType1:

POSITION vector, EZ[A1], below,
are the components of the 'Minkowski vector' $X \in O_{4,4}$
with respect to the vector basis of $O_{4,4}$:

parametrize each 7 - plane in terms of 7 free (independent) EZ[1], ..., EZ[7]
and a constrained Real part EZ[0] , assuming that it is not 0:

In[84]:=
```
Clear[eZ];
eZ = ToExpression["EZ[" <> ToString[#] <> "]"] & /@ Range[0, 7]
```
Out[85]=
```
{EZ[0], EZ[1], EZ[2], EZ[3], EZ[4], EZ[5], EZ[6], EZ[7]}
```

In[86]:=
```
eZ7 = ToExpression["Z7[" <> ToString[#] <> "]"] & /@ Range[6]
ssZ7 = %.Drop[Drop[eZ, 1], -1]
```
Out[86]=
```
{Z7[1], Z7[2], Z7[3], Z7[4], Z7[5], Z7[6]}
```
Out[87]=
```
EZ[1] × Z7[1] + EZ[2] × Z7[2] + EZ[3] × Z7[3] + EZ[4] × Z7[4] + EZ[5] × Z7[5] + EZ[6] × Z7[6]
```

In[88]:=
```
m6 = Drop[eZ, 1] /. {EZ[7] → ssZ7}
```
Out[88]=
```
{EZ[1], EZ[2], EZ[3], EZ[4], EZ[5], EZ[6],
 EZ[1] × Z7[1] + EZ[2] × Z7[2] + EZ[3] × Z7[3] + EZ[4] × Z7[4] + EZ[5] × Z7[5] + EZ[6] × Z7[6]}
```

In[89]:=
```
Dm6 = D[m6, #] & /@ Drop[Drop[eZ, 1], -1]
```
Out[89]=
```
{{1, 0, 0, 0, 0, 0, Z7[1]}, {0, 1, 0, 0, 0, 0, Z7[2]}, {0, 0, 1, 0, 0, 0, Z7[3]},
 {0, 0, 0, 1, 0, 0, Z7[4]}, {0, 0, 0, 0, 1, 0, Z7[5]}, {0, 0, 0, 0, 0, 1, Z7[6]}}
```

In[90]:=
```
Dimensions[Dm6]
```
Out[90]=
```
{6, 7}
```

In[91]:=
```
Block[{az, t},
 az = Array[a, 7];
 t = Table[Sum[
     kδ[Range[7], {a[1], a[2], a[3], a[4], a[5], a[6], a[7]}] *
      Dm6⟦1⟧⟦a[1]⟧ * Dm6⟦2⟧⟦a[2]⟧ *
      Dm6⟦3⟧⟦a[3]⟧ * Dm6⟦4⟧⟦a[4]⟧ * Dm6⟦5⟧⟦a[5]⟧ * Dm6⟦6⟧⟦a[6]⟧,
     {a[1], 1, 7}, {a[2], 1, 7}, {a[3], 1, 7}, {a[4], 1, 7}, {a[5], 1, 7}, {a[6], 1, 7}],
    {a[7], 1, Length[Dm6⟦1⟧]}] // Simplify;
 t]
```
Out[91]=
```
{-Z7[1], -Z7[2], -Z7[3], -Z7[4], -Z7[5], -Z7[6], 1}
```

In[92]:=
```
sztype2 = Solve[0 == vectorFormOfType2.eZ, EZ[0]][[-1]] // FullSimplify;
sevenPlane2 = eZ /. sztype2
```

Out[93]=
```
{((EZ[3] - EZ[4]) f16[9][x0, x1, x2, x3, x4, x5, x6, x7] +
    (-EZ[2] + EZ[5]) f16[10][x0, x1, x2, x3, x4, x5, x6, x7] +
    (EZ[1] + EZ[6]) f16[11][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[7] (-f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
       f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
    (EZ[3] + EZ[4]) f16[13][x0, x1, x2, x3, x4, x5, x6, x7] -
    (EZ[2] + EZ[5]) f16[14][x0, x1, x2, x3, x4, x5, x6, x7] +
    (EZ[1] - EZ[6]) f16[15][x0, x1, x2, x3, x4, x5, x6, x7]) /
  (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]),
 EZ[1], EZ[2], EZ[3],
 EZ[4],
 EZ[5],
 EZ[6],
 EZ[7]}
```

In[94]:=
```
sztype1 = Solve[0 == vectorFormOfType1.eZ, EZ[0]][[-1]] // (*Full*)Simplify;
sevenPlane1 = eZ /. sztype1
```

Out[95]=
```
{(-EZ[7] × f16[0][x0, x1, x2, x3, x4, x5, x6, x7] +
    (EZ[3] - EZ[4]) f16[1][x0, x1, x2, x3, x4, x5, x6, x7] -
    EZ[2] × f16[2][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[5] × f16[2][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[1] × f16[3][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[6] × f16[3][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[7] × f16[4][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[3] × f16[5][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[4] × f16[5][x0, x1, x2, x3, x4, x5, x6, x7] -
    EZ[2] × f16[6][x0, x1, x2, x3, x4, x5, x6, x7] -
    EZ[5] × f16[6][x0, x1, x2, x3, x4, x5, x6, x7] +
    EZ[1] × f16[7][x0, x1, x2, x3, x4, x5, x6, x7] -
    EZ[6] × f16[7][x0, x1, x2, x3, x4, x5, x6, x7]) /
  (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]),
 EZ[1], EZ[2], EZ[3],
 EZ[4],
 EZ[5],
 EZ[6],
 EZ[7]}
```

6 - plane :

In[96]:=
```
solve6Plane =
  Solve[And @@ Thread[sevenPlane2 == sevenPlane1], EZ[7]][[-1]] // (*Full*)Simplify;
sixPlaneAlpha = Drop[eZ, 1] /. solve6Plane
```

Out[97]=

$$\{EZ[1], EZ[2], EZ[3], EZ[4], EZ[5], EZ[6],$$

$$\Big((f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])$$

$$(f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7])$$

$$\Big(\frac{(EZ[3] - EZ[4])\, f16[1][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{EZ[2] \times f16[2][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[5] \times f16[2][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[1] \times f16[3][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[6] \times f16[3][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[3] \times f16[5][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[4] \times f16[5][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{EZ[2] \times f16[6][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{EZ[5] \times f16[6][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{EZ[1] \times f16[7][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{EZ[6] \times f16[7][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{(EZ[3] - EZ[4])\, f16[9][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{(EZ[2] - EZ[5])\, f16[10][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{(EZ[1] + EZ[6])\, f16[11][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

$$\frac{(EZ[3] + EZ[4])\, f16[13][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]} +$$

$$\frac{(EZ[2] + EZ[5])\, f16[14][x0, x1, x2, x3, x4, x5, x6, x7]}{f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]} -$$

```
                (EZ[1] - EZ[6]) f16[15][x0, x1, x2, x3, x4, x5, x6, x7]
            f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]  ))/
      (2 (-f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
          f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7]))}
```

In[98]:=
```
normalM6 = D[sixPlaneAlpha(*[[7]]*), #] & /@ Drop[Drop[eZ, 1], -1] // Simplify
```

Out[98]=
```
{{1, 0, 0, 0, 0, 0,
  (f16[3][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[7][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) -
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[11][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[15][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (-2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])},
 {0, 1, 0, 0, 0, 0, (f16[2][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1,
         x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[6][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) -
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[10][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[14][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] - 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])},
 {0, 0, 1, 0, 0, 0, (f16[1][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1,
         x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[5][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) -
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[9][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[13][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (-2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])},
 {0, 0, 0, 1, 0, 0, (-f16[1][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1,
         x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[5][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
        f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[9][x0, x1, x2, x3, x4, x5, x6, x7] -
        f16[13][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (-2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] + 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])},
```

```
{0, 0, 0, 0, 1, 0, (-f16[2][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1,
          x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[6][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
         f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[10][x0, x1, x2, x3, x4, x5, x6, x7] -
         f16[14][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] - 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])},
{0, 0, 0, 0, 0, 1, (-f16[3][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1,
          x2, x3, x4, x5, x6, x7] + f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      f16[7][x0, x1, x2, x3, x4, x5, x6, x7] (f16[8][x0, x1, x2, x3, x4, x5, x6, x7] +
         f16[12][x0, x1, x2, x3, x4, x5, x6, x7]) +
      (f16[0][x0, x1, x2, x3, x4, x5, x6, x7] + f16[4][x0, x1, x2, x3, x4, x5, x6, x7])
       (f16[11][x0, x1, x2, x3, x4, x5, x6, x7] -
         f16[15][x0, x1, x2, x3, x4, x5, x6, x7])) /
    (2 f16[4][x0, x1, x2, x3, x4, x5, x6, x7] × f16[8][x0, x1, x2, x3, x4, x5, x6, x7] - 2
       f16[0][x0, x1, x2, x3, x4, x5, x6, x7] × f16[12][x0, x1, x2, x3, x4, x5, x6, x7])}}
```

In[99]:=

```
(*makeSevenPlane2Real=
  Solve[0==EZ[0]/.sztype2,EZ[7](*Complement[eZ,{EZ[0]}]*)]//Simplify
 sixPlane2=eZ/.Flatten[{{EZ[0]→0},makeSevenPlane2Real}]//Simplify*)
```

In[100]:=

```
(*makeSevenPlane1Real=
  Solve[0==EZ[0]/.sztype1,EZ[7](*Complement[eZ,{EZ[0]}]*)]//Simplify
 sixPlane1=eZ/.Flatten[{{EZ[0]→0},makeSevenPlane1Real}]//Simplify*)
```

In[101]:=

```
Complement[eZ, {EZ[0], EZ[7]}]
```

Out[101]=

```
{EZ[1], EZ[2], EZ[3], EZ[4], EZ[5], EZ[6]}
```

In[102]:=

```
(*sixPlane=
 Solve[0==(EZ[0]/.real1)&&0==(EZ[0]/.real2),Complement[eZ,{EZ[0],EZ[7]}]][[-1]]//
  (*Full*)Simplify*)
```

Assuming that sevenPlane2 is not parallel to sevenPlane1,
find the 6-plane of their intersection:

In[103]:=

```
(*sixPlane=
 Solve[And@@Thread[sevenPlane1==sevenPlane2],Complement[eZ,{EZ[0]}]][[-1]]*)
```

parametrize the 6-plane in terms of 6 free (independent) EZ[1], ..., EZ[6]
and constrained EZ[7], assuming that it is not 0:

two 7 - planes : intersection, if not parallel, is 6-plane M^6
remember, only one split octonion algebra $\mathbb{O}_{4,4}$, with three distinct bases.

Spinor Lagrangian

In[104]:=
```
Protect[H, M];
```

In[105]:=
```
constraintVars = {};
```

In[106]:=
```
Clear[lagrangian16];
lagrangian16[] := 1/2 (Transpose[Ψ16].σ1616.
      Sum[CA8[α1 - 1].(D[Ψ16, X⟦α1⟧]), {α1, 1, Length[X]}] +
     M * Transpose[Ψ16].σ1616.Ψ16) // Simplify[#, constraintVars] &
```

check Transpose[lagrangian16[]] === lagrangian16[] :

In[108]:=
```
2 lagrangian16[];
- Sum[(D[Transpose[Ψ16], X⟦α1⟧]).σ1616.CA8[α1 - 1], {α1, 1, Length[X]}].
     Ψ16 + (M) * Transpose[Ψ16].σ1616.Ψ16 //
  (*Full*)Simplify[#, constraintVars] &;
ExpandAll[% - %%] // (*Full*)Simplify[#, constraintVars] &
```

Out[110]=
```
0
```

recheck Transpose[lagrangian16[]] === lagrangian16[] :

In[111]:=
```
(Transpose[Ψ16].σ1616.Sum[CA8[α1 - 1].(D[Ψ16, X⟦α1⟧]), {α1, 1, Length[X]}] +
    (M) * Transpose[Ψ16].σ1616.Ψ16) // (*Full*)Simplify[#, constraintVars] &;
- Sum[(D[Transpose[Ψ16], X⟦α1⟧]).σ1616.CA8[α1 - 1], {α1, 1, Length[X]}].Ψ16 +
   (M) * Transpose[Ψ16].σ1616.Ψ16 // (*Full*)Simplify[#, constraintVars] &;
ExpandAll[% - %%] // (*Full*)Simplify[#, constraintVars] &
```

Out[113]=
```
0
```

In[114]:=

```
Sum[ Block[{x, y, p, t, F = Ψ16⟦5⟧},
  x = X⟦α1⟧;
  y = D[F, x];
  p = D[lagrangian16[], y];
  t = D[p, x];
  t], {α1, 1, Length[X]}]
```

Out[114]=

$$\frac{1}{2} f16[8]^{(0,0,0,0,0,0,0,1)}[x0, x1, x2, x3, x4, x5, x6, x7] + \frac{1}{2} f16[15]^{(0,0,0,0,0,0,1,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \frac{1}{2} f16[14]^{(0,0,0,0,0,1,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \frac{1}{2} f16[13]^{(0,0,0,0,1,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \frac{1}{2} f16[13]^{(0,0,0,1,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \frac{1}{2} f16[14]^{(0,0,1,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \frac{1}{2} f16[15]^{(0,1,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \frac{1}{2} f16[8]^{(1,0,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7]$$

In[115]:=

```
Table[Block[{F, s}, F = Ψ16⟦k⟧;
  s = Reap[For[A1 = 0, A1 < 8, A1++, Sow[Block[{x, y, p, t},
        x = X⟦1 + A1⟧;
        y = D[F, x];
        p = D[lagrangian16[], y];
        t = D[p, x];
        t]
      ]]] // Drop[#, 1] &;
  s]
 , {k, 1, Length[Ψ16] / 8}]
```

Out[115]=

$$\left\{\left\{\left\{\left\{\frac{1}{2}\,\mathrm{f16}[12]^{(1,0,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],\right.\right.\right.\right.$$
$$-\frac{1}{2}\,\mathrm{f16}[11]^{(0,1,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\frac{1}{2}\,\mathrm{f16}[10]^{(0,0,1,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[9]^{(0,0,0,1,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\frac{1}{2}\,\mathrm{f16}[9]^{(0,0,0,0,1,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[10]^{(0,0,0,0,0,1,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[11]^{(0,0,0,0,0,0,1,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\left.\left.\left.-\frac{1}{2}\,\mathrm{f16}[12]^{(0,0,0,0,0,0,0,1)}[x0, x1, x2, x3, x4, x5, x6, x7]\right\}\right\}\right\},$$
$$\left\{\left\{\left\{\frac{1}{2}\,\mathrm{f16}[13]^{(1,0,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],\right.\right.\right.$$
$$-\frac{1}{2}\,\mathrm{f16}[10]^{(0,1,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[11]^{(0,0,1,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\frac{1}{2}\,\mathrm{f16}[8]^{(0,0,0,1,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[8]^{(0,0,0,0,1,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$-\frac{1}{2}\,\mathrm{f16}[11]^{(0,0,0,0,0,1,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\frac{1}{2}\,\mathrm{f16}[10]^{(0,0,0,0,0,0,1,0)}[x0, x1, x2, x3, x4, x5, x6, x7],$$
$$\left.\left.\left.\left.-\frac{1}{2}\,\mathrm{f16}[13]^{(0,0,0,0,0,0,0,1)}[x0, x1, x2, x3, x4, x5, x6, x7]\right\}\right\}\right\}\right\}$$

Euler-Lagrange Expression

In[116]:=

```
Clear[eulerLagrangeExpression];
eulerLagrangeExpression[] := Module[{L, t},
  L = lagrangian16[];
  t = Table[Block[{F, s}, F = Ψ16⟦k⟧;
      s = D[L, F] - Sum[ Block[{x, y, p, q},
          x = X⟦α1⟧;
          y = D[F, x];
          p = D[L, y];
          q = D[p, x];
          q], {α1, 1, Length[X]}];
      s]
     , {k, 1, Length[Ψ16]}] // FullSimplify;
  Return[t];
 ]
```

eLa

In[118]:=

```
(*Get[
  "2025-12-14_Q2=0-WaveFunctionUniverse-4+4-Einstein-Lovelock-Nash-L9i-eLa.
    mx"];*)
```

In[119]:=

```
eLa = eulerLagrangeExpression[];
```

In[120]:=

```
eLa // Column
```

Out[120]=

$$
\begin{aligned}
&-M\, f16[4][x0, x1, x2, x3, x4, x5, x6, x7] + \\
&f16[12]^{(0,0,0,0,0,0,0,1)}[x0, x1, x2, x3, x4, x5, x6, x7] + \\
&f16[11]^{(0,0,0,0,0,0,1,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \\
&f16[10]^{(0,0,0,0,0,1,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \\
&f16[9]^{(0,0,0,0,1,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \\
&f16[9]^{(0,0,0,1,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \\
&f16[10]^{(0,0,1,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] + \\
&f16[11]^{(0,1,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7] - \\
&f16[12]^{(1,0,0,0,0,0,0,0)}[x0, x1, x2, x3, x4, x5, x6, x7]
\end{aligned}
$$

```
-M f16[5][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[13]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[10]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[11]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[8]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[11]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[10]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[13]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
-M f16[6][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[14]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[9]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[11]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[11]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[8]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[9]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[14]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
-M f16[7][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[15]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[9]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[10]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[10]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[9]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[15]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
-M f16[0][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[15]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[14]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[13]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[13]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[14]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[15]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[8]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
```

```
-M f16[1][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[9]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[14]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[15]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[12]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[12]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[15]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[14]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[9]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
-M f16[2][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[10]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[13]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[12]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[15]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[15]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[12]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[13]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[10]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
-M f16[3][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[11]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[12]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[13]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[14]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[14]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[13]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[12]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[11]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[12][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[4]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[2]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[1]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[1]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[2]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[4]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
```

```
M f16[13][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[5]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[2]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[0]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[0]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[2]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[5]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[14][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[6]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[1]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[0]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[0]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[1]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[6]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[15][x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[7]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[0]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[1]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[2]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[2]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[1]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[0]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[7]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[8][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[0]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[7]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[6]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[5]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[5]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[6]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[7]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[0]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
```

```
M f16[9][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[1]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[6]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[7]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[4]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[4]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[7]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[6]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[1]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[10][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[2]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[5]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[4]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[7]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[7]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[4]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[5]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[2]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
M f16[11][x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[3]^(0,0,0,0,0,0,0,1)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[4]^(0,0,0,0,0,0,1,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[5]^(0,0,0,0,0,1,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[6]^(0,0,0,0,1,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[6]^(0,0,0,1,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[5]^(0,0,1,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] -
 f16[4]^(0,1,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7] +
 f16[3]^(1,0,0,0,0,0,0,0)[x0, x1, x2, x3, x4, x5, x6, x7]
```

complexification of the split - octonion algebra

employ complexification $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{O}_{4,4}$, $\mathbb{C} \not\subset \mathbb{O}_{4,4}$, $\mathbb{i} \in \mathbb{C}$:

In[121]:=

```
Clear[sf16Aa];
sfψ16Aa = Table[f16[j] → ToExpression["((Q[" <> ToString[j] <>
      "]ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7))&)"], {j, 0, 15}]
```

Out[122]=

```
{f16[0] → (Q[0] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[1] → (Q[1] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[2] → (Q[2] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[3] → (Q[3] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[4] → (Q[4] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[5] → (Q[5] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[6] → (Q[6] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[7] → (Q[7] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[8] → (Q[8] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[9] → (Q[9] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[10] → (Q[10] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[11] → (Q[11] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[12] → (Q[12] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[13] → (Q[13] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[14] → (Q[14] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &),
 f16[15] → (Q[15] ⅇ^(ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)) &)}
```

← vector basis of $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{O}_{4,4}$, $\mathbb{i} \in \mathbb{C}$, $\mathbb{C} \not\subset \mathbb{O}_{4,4}$

The complexification of the split octonions results in the same algebra as the complexification of the standard octonions .

This unified structure is simply known as the complex octonions $\mathbb{O}_{\mathbb{C}}$.

The standard basis for the vector split – octonions shall be

$\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$, where $e_0 = 1$.

In the split case, the squares of the imaginary units typically have the signature

{+1, +1, +1, -1, -1, -1, -1, -1 }.

Introduce a new imaginary unit i, where $i^2 = -e_0 = -1$,

that commutes with all of the split - octonion basis elements e_A, $A = 0, 1, \ldots, 7$.

Define the new algebra as the set of all elements :

$$\mathbb{Z} = \sum_{A=0}^{7} z_A e_A \quad ,$$

where each z_A is a complex number $z_A = x_A + i\, y_A \in \mathbb{C}$.

Extend Multiplication : Define multiplication by requiring it to be $\mathbb{C}$ - linear .

If

$\mathbb{A}, \mathbb{B} \in \mathbb{O}_{\mathbb{C}}$,

where

$\mathbb{A} = \sum_{A=0}^{7} z_A \, e_A$,

$\mathbb{B} = \sum_{B=0}^{7} z_B \, e_B$,

then

$(z_A \, e_A)(z_B \, e_B) = (z_A \, z_B)(e_A \, e_B)$

In[123]:=

```
sfψ16Aa // Column
```

Out[123]=

```
f16[0] → (Q[0] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[1] → (Q[1] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[2] → (Q[2] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[3] → (Q[3] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[4] → (Q[4] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[5] → (Q[5] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[6] → (Q[6] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[7] → (Q[7] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[8] → (Q[8] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[9] → (Q[9] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[10] → (Q[10] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[11] → (Q[11] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[12] → (Q[12] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[13] → (Q[13] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[14] → (Q[14] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
f16[15] → (Q[15] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &)
```

In[124]:=

```
time1 = Now
AbsoluteTiming[
 (eLb = eLa / (ⅇ^ⅈ (k1 x0+k2 x1+k3 x2+k4 x3+k5 x4+k6 x5+k7 x6)) /. sfψ16Aa // FullSimplify) // Column]
Now - time1
```

Out[124]=

Mon 30 Mar 2026 06:26:07 GMT−7

Out[125]=

```
-ⅇ^ⅈ k0 (Q[4] ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) &) M Q[4] +
 ⅈ ⅇ^ⅈ k0 (Q[9] (ⅈ ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4) &) k4 Q[9] -
 ⅈ ⅇ^ⅈ k0 (Q[9] (ⅈ ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5) &) k5 Q[9] -
 ⅈ ⅇ^ⅈ k0 (Q[10] (ⅈ ⅇ^ⅈ (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3) &) k3 Q[10] +
```

```
i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[10] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[11] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[11] -
i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[12]
- e^(i k0 (Q[5] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[5] -
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[8] +
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[8] +
i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[10] -
i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[10] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[11] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[11] -
i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[13]
- e^(i k0 (Q[6] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[6] +
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[8] -
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[8] -
i e^(i k0 (Q[9] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[9] +
i e^(i k0 (Q[9] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[9] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[11] +
i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[11] -
i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[14]
- e^(i k0 (Q[7] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[7] -
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[8] -
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[8] -
i e^(i k0 (Q[9] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[9] -
i e^(i k0 (Q[9] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[9] -
i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[10] -
i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[10] -
i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[15]
- e^(i k0 (Q[0] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[0] -
i e^(i k0 (Q[8] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[8] +
i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[13] +
i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[13] -
i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[14] -
i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[14] +
i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[15] -
i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[15]
- e^(i k0 (Q[1] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[1] -
i e^(i k0 (Q[9] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[9] -
i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[12] -
i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[12] +
i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[14] +
```

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  i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[14] +
  i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[15] -
  i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[15]
 - e^(i k0 (Q[2] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[2] -
  i e^(i k0 (Q[10] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[10] +
  i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[12] +
  i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[12] -
  i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[13] -
  i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[13] +
  i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[15] -
  i e^(i k0 (Q[15] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[15]
 - e^(i k0 (Q[3] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[3] -
  i e^(i k0 (Q[11] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[11] -
  i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[12] +
  i e^(i k0 (Q[12] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[12] -
  i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[13] +
  i e^(i k0 (Q[13] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[13] -
  i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[14] +
  i e^(i k0 (Q[14] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[14]
 i (e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[1] -
      e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[1] -
      e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[2] +
      e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[2] +
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[3] +
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[3] +
      e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[4]) +
  e^(i k0 (Q[12] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[12]
 - i (e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[0] -
      e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[0] -
      e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[2] +
      e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[2] -
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[3] -
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[3] -
      e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&)) k1 Q[5]) +
  e^(i k0 (Q[13] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&)) M Q[13]
 i (e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&)) k3 Q[0] -
      e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&)) k6 Q[0] -
      e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&)) k2 Q[1] +
      e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&)) k7 Q[1] +
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&)) k4 Q[3] +
      e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&)) k5 Q[3] +
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        e^(i k0 (Q[6] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[6]) +
  e^(i k0 (Q[14] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&) M Q[14]
- i (e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&) k2 Q[0] +
        e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&) k7 Q[0] +
        e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&) k3 Q[1] +
        e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&) k6 Q[1] +
        e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&) k4 Q[2] +
        e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&) k5 Q[2] -
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[7]) +
  e^(i k0 (Q[15] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&) M Q[15]
i (e^(i k0 (Q[0] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[0] +
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&) k4 Q[5] +
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&) k5 Q[5] -
        e^(i k0 (Q[6] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&) k3 Q[6] -
        e^(i k0 (Q[6] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&) k6 Q[6] +
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&) k2 Q[7] -
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&) k7 Q[7]) +
  e^(i k0 (Q[8] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&) M Q[8]
i (e^(i k0 (Q[1] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[1] -
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&) k4 Q[4] -
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&) k5 Q[4] +
        e^(i k0 (Q[6] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&) k2 Q[6] +
        e^(i k0 (Q[6] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&) k7 Q[6] +
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&) k3 Q[7] -
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&) k6 Q[7]) +
  e^(i k0 (Q[9] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&) M Q[9]
i (e^(i k0 (Q[2] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[2] +
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&) k3 Q[4] +
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&) k6 Q[4] -
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&) k2 Q[5] -
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&) k7 Q[5] +
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k4)&) k4 Q[7] -
        e^(i k0 (Q[7] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k5)&) k5 Q[7]) +
  e^(i k0 (Q[10] e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7)&) M Q[10]
i (e^(i k0 (Q[3] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k1)&) k1 Q[3] -
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k2)&) k2 Q[4] +
        e^(i k0 (Q[4] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k7)&) k7 Q[4] -
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k3)&) k3 Q[5] +
        e^(i k0 (Q[5] (i e^(i (k0 #0+k1 #1+k2 #2+k3 #3+k4 #4+k5 #5+k6 #6+k7 #7) k6)&) k6 Q[5] -
```

$$e^{i\,k0}\left(Q[6]\left(i\,e^{i\,(k0\,\#0+k1\,\#1+k2\,\#2+k3\,\#3+k4\,\#4+k5\,\#5+k6\,\#6+k7\,\#7)}\,k4\right)\&\right)\,k4\,Q[6]\,+$$
$$e^{i\,k0}\left(Q[6]\left(i\,e^{i\,(k0\,\#0+k1\,\#1+k2\,\#2+k3\,\#3+k4\,\#4+k5\,\#5+k6\,\#6+k7\,\#7)}\,k5\right)\&\right)\,k5\,Q[6]\Big)\,+$$
$$e^{i\,k0}\left(Q[11]\,e^{i\,(k0\,\#0+k1\,\#1+k2\,\#2+k3\,\#3+k4\,\#4+k5\,\#5+k6\,\#6+k7\,\#7)}\&\right)\,M\,Q[11]$$

Out[126]=

0.912212 s

In[127]:=

```
Dimensions[eLb]
```

Out[127]=

{16}

In[128]:=

```
(*DumpSave[ToString[header<>"eLa.mx"],eLa];*)
```

In[129]:=

```
(*Print[ToString[header <> "eLa.mx"]]*)
```

In[130]:=

```
(*Cases[
    eLa,
    h_Symbol[n_Integer] /; StringEndsQ[SymbolName[h], "f16"] :> n,
    Infinity,
    Heads → True
]*)
```

WRITE OUT THE VARIABLES THAT ARE COUPLED TO ANOTHER VARIABLE:

In[131]:=

```
eLaRawSets = Cases[
  #,
  h_Symbol[n_Integer] /; StringEndsQ[SymbolName[h], "f16"] :> n,
  Infinity,
  Heads → True
] & /@ eLa
```

Out[131]=

```
{{4, 12, 11, 10, 9, 9, 10, 11, 12}, {5, 13, 10, 11, 8, 8, 11, 10, 13},
 {6, 14, 9, 8, 11, 11, 8, 9, 14}, {7, 15, 8, 9, 10, 10, 9, 8, 15},
 {0, 8, 15, 14, 13, 13, 14, 15, 8}, {1, 9, 14, 15, 12, 12, 15, 14, 9},
 {2, 10, 13, 12, 15, 15, 12, 13, 10}, {3, 11, 12, 13, 14, 14, 13, 12, 11},
 {12, 4, 3, 2, 1, 1, 2, 3, 4}, {13, 5, 2, 3, 0, 0, 3, 2, 5},
 {14, 6, 1, 0, 3, 3, 0, 1, 6}, {15, 7, 0, 1, 2, 2, 1, 0, 7}, {8, 0, 7, 6, 5, 5, 6, 7, 0},
 {9, 1, 6, 7, 4, 4, 7, 6, 1}, {10, 2, 5, 4, 7, 7, 4, 5, 2}, {11, 3, 4, 5, 6, 6, 5, 4, 3}}
```

In[132]:=

```
Dimensions[eLaRawSets]
```

Out[132]=

{16, 9}

In[133]:=

```
rawSetselQ16 = rawSets[eLb, "Q", 0]
```

Out[133]=

```
{{4, 4, 9, 9, 9, 9, 10, 10, 10, 10, 11, 11, 11, 11, 12, 12},
 {5, 5, 8, 8, 8, 8, 10, 10, 10, 10, 11, 11, 11, 11, 13, 13},
 {6, 6, 8, 8, 8, 8, 9, 9, 9, 9, 11, 11, 11, 11, 14, 14},
 {7, 7, 8, 8, 8, 8, 9, 9, 9, 9, 10, 10, 10, 10, 15, 15},
 {0, 0, 8, 8, 13, 13, 13, 13, 14, 14, 14, 14, 15, 15, 15, 15},
 {1, 1, 9, 9, 12, 12, 12, 12, 14, 14, 14, 14, 15, 15, 15, 15},
 {2, 2, 10, 10, 12, 12, 12, 12, 13, 13, 13, 13, 15, 15, 15, 15},
 {3, 3, 11, 11, 12, 12, 12, 12, 13, 13, 13, 13, 14, 14, 14, 14},
 {1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 12, 12},
 {0, 0, 0, 0, 2, 2, 2, 2, 3, 3, 3, 3, 5, 5, 13, 13},
 {0, 0, 0, 0, 1, 1, 1, 1, 3, 3, 3, 3, 6, 6, 14, 14},
 {0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 7, 7, 15, 15},
 {0, 0, 5, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8},
 {1, 1, 4, 4, 4, 4, 6, 6, 6, 6, 7, 7, 7, 7, 9, 9},
 {2, 2, 4, 4, 4, 4, 5, 5, 5, 5, 7, 7, 7, 7, 10, 10},
 {3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 11, 11}}
```

In[134]:=

```
Block[{i = 1, j = -1}, Delete[rawSetselQ16, {{i}, {j}}]]
```

Out[134]=

```
{{5, 5, 8, 8, 8, 8, 10, 10, 10, 10, 11, 11, 11, 11, 13, 13},
 {6, 6, 8, 8, 8, 8, 9, 9, 9, 9, 11, 11, 11, 11, 14, 14},
 {7, 7, 8, 8, 8, 8, 9, 9, 9, 9, 10, 10, 10, 10, 15, 15},
 {0, 0, 8, 8, 13, 13, 13, 13, 14, 14, 14, 14, 15, 15, 15, 15},
 {1, 1, 9, 9, 12, 12, 12, 12, 14, 14, 14, 14, 15, 15, 15, 15},
 {2, 2, 10, 10, 12, 12, 12, 12, 13, 13, 13, 13, 15, 15, 15, 15},
 {3, 3, 11, 11, 12, 12, 12, 12, 13, 13, 13, 13, 14, 14, 14, 14},
 {1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 12, 12},
 {0, 0, 0, 0, 2, 2, 2, 2, 3, 3, 3, 3, 5, 5, 13, 13},
 {0, 0, 0, 0, 1, 1, 1, 1, 3, 3, 3, 3, 6, 6, 14, 14},
 {0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 7, 7, 15, 15},
 {0, 0, 5, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8},
 {1, 1, 4, 4, 4, 4, 6, 6, 6, 6, 7, 7, 7, 7, 9, 9},
 {2, 2, 4, 4, 4, 4, 5, 5, 5, 5, 7, 7, 7, 7, 10, 10}}
```

In[135]:=

```
rawSetsel16 = rawSets[eLa, "f16", 0]
% === eLaRawSets
```

Out[135]=

```
{{4, 12, 11, 10, 9, 9, 10, 11, 12}, {5, 13, 10, 11, 8, 8, 11, 10, 13},
 {6, 14, 9, 8, 11, 11, 8, 9, 14}, {7, 15, 8, 9, 10, 10, 9, 8, 15},
 {0, 8, 15, 14, 13, 13, 14, 15, 8}, {1, 9, 14, 15, 12, 12, 15, 14, 9},
 {2, 10, 13, 12, 15, 15, 12, 13, 10}, {3, 11, 12, 13, 14, 14, 13, 12, 11},
 {12, 4, 3, 2, 1, 1, 2, 3, 4}, {13, 5, 2, 3, 0, 0, 3, 2, 5},
 {14, 6, 1, 0, 3, 3, 0, 1, 6}, {15, 7, 0, 1, 2, 2, 1, 0, 7}, {8, 0, 7, 6, 5, 5, 6, 7, 0},
 {9, 1, 6, 7, 4, 4, 7, 6, 1}, {10, 2, 5, 4, 7, 7, 4, 5, 2}, {11, 3, 4, 5, 6, 6, 5, 4, 3}}
```

Out[136]=

```
True
```

In[137]:=

```
eLaCouplings = showCoupledEquations[eLaRawSets]
% // Flatten // Sort
% === Range[0, 15]
```

Out[137]=

```
{{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}}
```

Out[138]=

```
{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}
```

Out[139]=

```
True
```

In[140]:=

```
eLaztCouplings = eLaCouplings
```

Out[140]=

```
{{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}}
```

Section 1: Recapitulation of sixPlaneAlpha

The ambient space is R^8 endowed with the SO(4,4)-invariant split-signature metric:

η88 = diag(1,1,1,1,-1,-1,-1,-1), i.e. <x,x> = x0^2+x1^2+x2^2+x3^2 - x4^2-x5^2-x6^2-x7^2.

This is the standard pseudo-Euclidean norm on R^{4,4} studied in the repository.

The 16-component Dirac spinor Psi = {Psi1, Psi2} is decomposed under the 8x8 chirality operator σ88 = γ88[1].γ88[2].γ88[3]:
* Psi1 (Type 1, left-handed): σ88 . Psi1 = +Psi1
* Psi2 (Type 2, right-handed): σ88 . Psi2 = -Psi2

Via Cartan's triality (which identifies the vector, Type-1 spinor, and Type-2 spinor representations of

Spin(4,4)), each 8-component chiral spinor maps to a vector in V = R^8:
v1 = triality(Psi1), v2 = triality(Psi2).

Two 7-planes (hyperplanes) are defined as the orthogonal complements to these vectors with respect to η88:
H1 = { x in V : <x, v1>_η88 = 0 } (dimension 7)
H2 = { x in V : <x, v2>_η88 = 0 } (dimension 7)

The intersection sixPlaneAlpha = H1 intersect H2 generically has dimension 8 - 2 = 6.

From the README: 'the 6-plane of their intersection identified; todo: associate 6-plane with 3 generations of Einstein-Rosen 2-planes.'

The key: 6 = 3 x 2, so sixPlaneAlpha decomposes into 3 real 2-planes, each one an Einstein-Rosen bridge generation.

Section 2: Explicit Construction of sixPlaneAlpha

We work with the ambient R^8 and split metric η88 = diag(1,1,1,1,-1,-1,-1,-1). The two 7-planes are defined by linear conditions on x: η88.v1 and η88.v2 are the normal covectors. sixPlaneAlpha = NullSpace of the 2x8 matrix [η88.v1; η88.v2]^T.

In[141]:=

```
(* ---- Symbolic construction ---- *)

(* Split metric for R^{4,4} *)
(*eta = DiagonalMatrix[{1,1,1,1,-1,-1,-1,-1}];*)
(*eta===η88*)
Print["'Ambient' split metric η88 = ", MatrixForm[η88]];
```

'Ambient' split metric η88 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

In[142]:=

```
(* Symbolic normal vectors v1, v2 in R^8 (image of chiral spinors under triality) *)
Clear[v1s,v1,v2s,v2]
v1s = Array[Subscript[v1,#]&, 8];
v2s = Array[Subscript[v2,#]&, 8];

(* The two 7-planes H1, H2 are defined by the vanishing of η88-inner products *)
(* H1 = {x : x.η88.v1 = 0},  H2 = {x : x.η88.v2 = 0} *)
(* Normal covectors in the dual space *)
n1 = η88 . v1s;
n2 = η88 . v2s;

(* sixPlaneAlpha = H1 intersect H2 = NullSpace of normalMatrix *)
normalMatrix = {n1, n2};
Print["Normal matrix (2x8) whose null space is sixPlaneAlpha:"];
Print[MatrixForm[normalMatrix]];
```

Normal matrix (2x8) whose null space is sixPlaneAlpha:

$$\begin{pmatrix} v1_1 & v1_2 & v1_3 & v1_4 & -v1_5 & -v1_6 & -v1_7 & -v1_8 \\ v2_1 & v2_2 & v2_3 & v2_4 & -v2_5 & -v2_6 & -v2_7 & -v2_8 \end{pmatrix}$$

In[159]:=

```
(* ---- Numerical example ---- *)

(* Choose explicit unit vectors v1, v2 in R^8 (generic position) *)
(* v1 and v2 should be independent and have non-zero η88-norms *)
v1num = {1,0,0,0,0,0,0,0};   (* e0 direction: η88-norm +1 *)
v2num = {0,1,0,0,0,0,0,0};   (* e1 direction: η88-norm +1 *)

n1num = η88 . v1num;
n2num = η88 . v2num;
normalMatrixNum = {n1num, n2num};

(* sixPlaneAlpha as explicit 6x8 basis matrix (rows are basis vectors) *)
sixPlaneAlphaBasis = NullSpace[normalMatrixNum];
Print["sixPlaneAlpha basis (" <> ToString[Length[sixPlaneAlphaBasis]] <> " vectors in R^8)
Print[MatrixForm[sixPlaneAlphaBasis]];
Print["Dimension of sixPlaneAlpha: ", Length[sixPlaneAlphaBasis]];
```

sixPlaneAlpha basis (6 vectors in R^8):

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Dimension of sixPlaneAlpha: 6

In[175]:=

```
(* ---- Verify orthogonality of basis to normal vectors ---- *)

Print["Verification - basis vectors are orthogonal to n1num:"];
Print[MatrixForm[sixPlaneAlphaBasis . n1num]];
Print["Verification - basis vectors are orthogonal to n2num:"];
Print[MatrixForm[sixPlaneAlphaBasis . n2num]];
```

Verification - basis vectors are orthogonal to n1num:

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Verification - basis vectors are orthogonal to n2num:

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Section 3: What is a U(3) Structure?

A U(3) structure on a real 6-dimensional manifold M is a triple (g, J, omega) consisting of:

(1) A Riemannian metric g: a symmetric positive-definite bilinear form on TM.

(2) An almost-complex structure J: TM -> TM satisfying J^2 = -Id_{TM}. This makes M look locally like C^3.

(3) A Kähler (fundamental) 2-form omega: a non-degenerate antisymmetric bilinear form.

These three objects must satisfy the compatibility conditions:
(a) g(JX, JY) = g(X, Y) for all vector fields X, Y [J is an isometry of g]
(b) omega(X, Y) = g(JX, Y) [omega is determined by g and J]
(c) omega is non-degenerate [i.e., omega^3 is a volume form]

When all three conditions hold, the structure group of the frame bundle reduces from GL(6,R) to U(3) inside SO(6).

Concretely on R^6: identify R^6 with C^3 via the map
(x1,x2,x3,x4,x5,x6) <-> (x1+i*x4, x2+i*x5, x3+i*x6).
Then:
* The standard complex structure J acts as J(e_{2k-1}) = e_{2k}, J(e_{2k}) = -e_{2k-1}, k=1,2,3.

* The standard Kähler form is omega = dx1^dx4 + dx2^dx5 + dx3^dx6.
* U(3) = {A in GL(6,R) : A^T g A = g, A J = J A}.

This decomposition 6 = 2+2+2 is what relates the sixPlaneAlpha to 3 generations of Einstein-Rosen 2-planes.

Section 4: The Induced Metric on sixPlaneAlpha

The ambient space R^8 carries the split metric η88 = diag(1,1,1,1,-1,-1,-1,-1). The pullback of η88 to sixPlaneAlpha via the embedding i: sixPlaneAlpha -> R^8 gives an induced bilinear form.

In[181]:=

```
(* ---- Induced (pullback) metric on sixPlaneAlpha ---- *)

(* B = 6x8 basis matrix of sixPlaneAlpha; rows are basis vectors in R^8 *)
B = sixPlaneAlphaBasis;  (* 6 x 8 *)

(* Induced metric = B . η88 . Transpose[B],  a 6x6 matrix *)
gInduced = B . η88 . Transpose[B];
Print["Induced (pullback) bilinear form on sixPlaneAlpha:"];
Print[MatrixForm[gInduced]];
Print["Eigenvalues of induced form: ", Eigenvalues[gInduced]];
```

Induced (pullback) bilinear form on sixPlaneAlpha:

$$\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Eigenvalues of induced form: {-1, -1, -1, -1, 1, 1}

For a U(3) structure we need a positive-definite metric on the 6-plane. If the normal vectors v1, v2 have positive η88-norm (lie in the Riemannian subspace), then the induced metric on their orthogonal complement may be indefinite. We therefore choose the Euclidean (positive-definite) metric g = Identity on the abstract 6-plane, independent of the ambient split signature. This is the standard approach when defining a U(3) structure on a subspace.

In[191]:=

```
(* ---- Positive-definite metric for U(3) structure ---- *)

(* We define g as the standard Euclidean metric on the abstract 6-plane *)
(* (The basis vectors of sixPlaneAlpha are treated as an orthonormal frame) *)
dim6 = Length[B];
g6 = IdentityMatrix[dim6];  (* 6x6 positive-definite metric *)
Print["Positive-definite metric g on sixPlaneAlpha (6x6 identity):"];
Print[MatrixForm[g6]];
```

Positive-definite metric g on sixPlaneAlpha (6x6 identity):

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Section 5: The Almost-Complex Structure J on sixPlaneAlpha

We define J on the abstract 6-dimensional space R^6 = C^3 as the standard complex structure. In the ordered basis {e1,e2,e3,e4,e5,e6} = {Re(z1),Re(z2),Re(z3),Im(z1),Im(z2),Im(z3)}, J acts as:

J(e_k) = e_{k+3} for k=1,2,3
J(e_{k+3}) = -e_k for k=1,2,3

In block form: J = [[0, -I3], [I3, 0]] (as a 6x6 real matrix).

Equivalently, using the 2x2 block j = [[0,-1],[1,0]], we have J = BlockDiagonalMatrix[{j,j,j}] in the basis that interleaves real and imaginary parts: {Re(z1),Im(z1),Re(z2),Im(z2),Re(z3),Im(z3)}.

In[199]:=

```
(* ---- Define the almost-complex structure J ---- *)

(* 2x2 rotation by 90 degrees: the basic complex unit *)
j2 = {{0,-1},{1,0}};

(* J = block-diagonal with three copies of j2, acting on C^3 = R^6 *)
(* Basis ordering: {Re(z1),Im(z1), Re(z2),Im(z2), Re(z3),Im(z3)} *)
J6 = BlockDiagonalMatrix[{j2, j2, j2}];
Print["Almost-complex structure J on sixPlaneAlpha:"];
Print[MatrixForm[J6]];

(* Verify J^2 = -Identity *)
J6sq = J6 . J6;
Print["J^2 = ", If[J6sq == -IdentityMatrix[6], "-Identity (VERIFIED)", "ERROR: J^2 != -Ide
```

Almost-complex structure J on sixPlaneAlpha:

$$\begin{pmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

J^2 = -Identity (VERIFIED)

Section 6: The Kähler Form omega on sixPlaneAlpha

Given the metric g and the almost-complex structure J, the Kähler (fundamental) 2-form omega is defined by:

omega(X,Y) = g(JX, Y)

for any tangent vectors X, Y on sixPlaneAlpha. As a 6x6 matrix:
omega_{ij} = omega(e_i, e_j) = g(J e_i, e_j) = sum_k J_{ki} g_{kj} = (J^T . g)_{ij}.
With g = Identity this gives omega = J^T = -J (since j2^T = -j2).

In[213]:=

```
(* ---- Define the Kähler form omega ---- *)

(* omega_{ij} = g(J e_i, e_j) = sum_k J_{ki} g_{kj} = (J^T . g)_{ij} *)
(* With g = Identity: omega = J^T = -J (since j2^T = -j2) *)
omega6 = Transpose[J6] . g6;
Print["Kähler form omega on sixPlaneAlpha:"];
Print[MatrixForm[omega6]];

(* Verify antisymmetry: omega^T = -omega *)
Print["omega is antisymmetric: ", If[Transpose[omega6] == -omega6, "True (VERIFIED)", "ERR

(* Verify non-degeneracy: det(omega) != 0 *)
detΩ = Det[omega6];
Print["det(omega) = ", detΩ, "  (non-degenerate: ", If[detΩ != 0, "True", "False"], ")"];

(* The top power omega^3 should be a volume form (non-zero multiple of the volume element)
omega3 = omega6 . omega6 . omega6;
Print["omega^3 = ", If[Det[omega3] != 0, "non-zero (volume form VERIFIED)", "ERROR: degene
```

Kähler form omega on sixPlaneAlpha:

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}$$

omega is antisymmetric: True (VERIFIED)

det(omega) = 1 (non-degenerate: True)

omega^3 = non-zero (volume form VERIFIED)

Section 7: Compatibility Verification of U(3) Conditions

We verify all three U(3) compatibility conditions:
(a) g(JX, JY) = g(X, Y) [J preserves g]

(b) omega(X, Y) = g(JX, Y) [Kähler condition]

(c) omega is non-degenerate [already verified above]

In[231]:=

```
(* ---- Verify all U(3) compatibility conditions ---- *)

(* (a) J^T . g . J = g  <==>  J^T . J = Identity (since g = Identity) *)
condA = Transpose[J6] . g6 . J6;
Print["Condition (a): J^T g J = g: ", If[condA == g6, "True (VERIFIED)", "ERROR"]];

(* (b) omega(X,Y) = g(JX,Y). As a matrix: omega_{ij} = g(J e_i, e_j) = (J^T . g)_{ij} *)
(* With g = Identity: omega = J^T = -J *)
(* We verify omega6 (defined as J^T . g in Section 6) matches this convention *)
condB_matrix = Transpose[J6] . g6;  (* = J^T = -J *)
Print["Condition (b): omega = J^T . g: ", If[condB_matrix == omega6, "True (VERIFIED)", "E
Print["omega6 is antisymmetric: ", If[Transpose[omega6] == -omega6, "True", "False"]];
Print["det(omega6) = ", Det[omega6]];
```

```
Condition (a): J^T g J = g: True (VERIFIED)

Condition (b): omega = J^T . g:
 If[condB_matrix == {{0, 1, 0, 0, 0, 0}, {-1, 0, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0},
    {0, 0, -1, 0, 0, 0}, {0, 0, 0, 0, 0, 1}, {0, 0, 0, 0, -1, 0}}, True (VERIFIED), ERROR]

omega6 is antisymmetric: True

det(omega6) = 1
```

In[244]:=

```
(* ---- U(3) structure summary ---- *)

Print["\n=== U(3) STRUCTURE ON sixPlaneAlpha SUMMARY ==="];
Print["Metric g (6x6): ", If[g6 == IdentityMatrix[6], "Standard Euclidean metric", "Custom
Print["Almost-complex J (6x6): J^2 = -I_6: ", If[J6.J6 == -IdentityMatrix[6], "VERIFIED",
Print["Kähler form omega = J^T.g: antisymmetric: ", If[Transpose[omega6] == -omega6, "VERI
Print["All U(3) compatibility conditions: ", If[Transpose[J6].g6.J6 == g6, "VERIFIED", "ER
```

```
=== U(3) STRUCTURE ON sixPlaneAlpha SUMMARY ===

Metric g (6x6): Standard Euclidean metric

Almost-complex J (6x6): J^2 = -I_6: VERIFIED

Kähler form omega = J^T.g: antisymmetric: VERIFIED

All U(3) compatibility conditions: VERIFIED
```

Section 8: Decomposition into 3 Complex Lines = 3 Generations

The 6-dimensional real space sixPlaneAlpha carries the complex structure J, which gives it the structure of C^3. As a complex vector space, C^3 decomposes into 3 complex lines (each a copy of C^1 = R^2):

sixPlaneAlpha = E1 + E2 + E3 (complex orthogonal direct sum)

where each Ek is a J-invariant real 2-plane spanned by {e_{2k-1}, e_{2k}}:
E1 = span{e1, e2} = span{Re(z1), Im(z1)} (first complex line)
E2 = span{e3, e4} = span{Re(z2), Im(z2)} (second complex line)
E3 = span{e5, e6} = span{Re(z3), Im(z3)} (third complex line)

Each Ek is a real Einstein-Rosen 2-plane bridge, corresponding to one generation of the Standard Model. The identification:
E1 <-> first generation (electron, up/down quarks)
E2 <-> second generation (muon, charm/strange quarks)
E3 <-> third generation (tau, top/bottom quarks)

is exactly the decomposition sought in the README: '3 generations of Einstein-Rosen 2-planes'.

In[251]:=

```
(* ---- Explicit 2-plane basis vectors ---- *)

(* Standard basis of R^6 *)
e = IdentityMatrix[6];

(* Three J-invariant 2-planes *)
E1basis = {e[[1]], e[[2]]};  (* Re(z1), Im(z1) *)
E2basis = {e[[3]], e[[4]]};  (* Re(z2), Im(z2) *)
E3basis = {e[[5]], e[[6]]};  (* Re(z3), Im(z3) *)

Print["E1 (1st generation ER 2-plane basis):", MatrixForm[E1basis]];
Print["E2 (2nd generation ER 2-plane basis):", MatrixForm[E2basis]];
Print["E3 (3rd generation ER 2-plane basis):", MatrixForm[E3basis]];

(* Verify each Ek is J-invariant: J.ek1 = ek2,  J.ek2 = -ek1 *)
Print["J-invariance of E1: J.e1 = e2? ",
  If[J6.e[[1]] == e[[2]] && J6.e[[2]] == -e[[1]], "VERIFIED", "ERROR"]];
Print["J-invariance of E2: J.e3 = e4? ",
  If[J6.e[[3]] == e[[4]] && J6.e[[4]] == -e[[3]], "VERIFIED", "ERROR"]];
Print["J-invariance of E3: J.e5 = e6? ",
  If[J6.e[[5]] == e[[6]] && J6.e[[6]] == -e[[5]], "VERIFIED", "ERROR"]];

(* The full space decomposes as direct sum *)
allBasis = Join[E1basis, E2basis, E3basis];
Print["Direct sum E1 + E2 + E3 spans R^6: ",
  If[Det[allBasis] != 0, "VERIFIED (full rank)", "ERROR"]];
```

E1 (1st generation ER 2-plane basis): $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$

E2 (2nd generation ER 2-plane basis): $\begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$

E3 (3rd generation ER 2-plane basis): $\begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$

```
J-invariance of E1: J.e1 = e2? VERIFIED
J-invariance of E2: J.e3 = e4? VERIFIED
J-invariance of E3: J.e5 = e6? VERIFIED
Direct sum E1 + E2 + E3 spans R^6: VERIFIED (full rank)
```

Section 9: U(3) Lie Algebra Generators on sixPlaneAlpha

The Lie algebra u(3) = u(1) + su(3) acts on C^3 = sixPlaneAlpha by complex-linear transformations preserving the Hermitian metric. In the real 6x6 representation, u(3) consists of skew-symmetric matrices that commute with J:

u(3) = {A in gl(6,R) : A^T g + g A = 0, A J = J A}

The generators split as:
U(1): generated by J itself (the overall phase rotation)
SU(3): generated by the 8 Gell-Mann matrices lambda_1,...,lambda_8, embedded as real 6x6 matrices via C^3 -> R^6.

In[273]:=

```
(* ---- U(1) generator ---- *)

(* The U(1) generator is J itself (rotation by 90 degrees in each complex plane) *)
U1gen = J6;
Print["U(1) generator = J:"];
Print[MatrixForm[U1gen]];
Print["J commutes with J (trivially): ", If[J6.U1gen - U1gen.J6 == 0*J6, "True", "False"]]
```

U(1) generator = J:

$$\begin{pmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

J commutes with J (trivially): True

In[280]:=

```
(* ---- SU(3) generators: Gell-Mann matrices as real 6x6 matrices ---- *)

(* The Gell-Mann matrices act on C^3 = R^6.  We embed each complex 3x3 Gell-Mann matrix *)
(* as a real 6x6 matrix via the standard real representation of complex numbers:  *)
(*   a + i*b  ->  {{a,-b},{b,a}}                                                   *)
(* A complex 3x3 matrix M = Re(M) + i*Im(M) maps to the real 6x6 matrix:          *)
(*   realRep[M] = KroneckerProduct[Re(M), {{1,0},{0,1}}]                           *)
(*              + KroneckerProduct[Im(M), {{0,-1},{1,0}}]                           *)
(*              = KroneckerProduct[Re(M), ID2] + KroneckerProduct[Im(M), j2]         *)

realRep6[M_] := KroneckerProduct[Re[M], IdentityMatrix[2]] +
                KroneckerProduct[Im[M], j2];

(* Complex 3x3 Gell-Mann matrices *)
lam1c = {{0,1,0},{1,0,0},{0,0,0}};
lam2c = {{0,-I,0},{I,0,0},{0,0,0}};
lam3c = {{1,0,0},{0,-1,0},{0,0,0}};
lam4c = {{0,0,1},{0,0,0},{1,0,0}};
lam5c = {{0,0,-I},{0,0,0},{I,0,0}};
lam6c = {{0,0,0},{0,0,1},{0,1,0}};
lam7c = {{0,0,0},{0,0,-I},{0,I,0}};
lam8c = (1/Sqrt[3])*{{1,0,0},{0,1,0},{0,0,-2}};

(* Real 6x6 embeddings *)
lam = {realRep6[lam1c], realRep6[lam2c], realRep6[lam3c],
       realRep6[lam4c], realRep6[lam5c], realRep6[lam6c],
       realRep6[lam7c], realRep6[lam8c]};

Print["Gell-Mann lambda_1 (real 6x6 embedding):"];
Print[MatrixForm[lam[[1]]]];
Print["Gell-Mann lambda_2 (real 6x6 embedding):"];
Print[MatrixForm[lam[[2]]]];
Print["Gell-Mann lambda_8 (real 6x6 embedding):"];
Print[MatrixForm[lam[[8]]]];
```

Gell-Mann lambda_1 (real 6x6 embedding):

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Gell-Mann lambda_2 (real 6x6 embedding):

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Gell-Mann lambda_8 (real 6x6 embedding):

$$\begin{pmatrix} \frac{1}{\sqrt{3}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2}{\sqrt{3}} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2}{\sqrt{3}} \end{pmatrix}$$

In[311]:=

```
(* ---- Verify all Gell-Mann generators commute with J ---- *)

Print["Verifying all lambda_k commute with J (required for u(3) action on C^3):"];
Do[
  comm = lam[[k]] . J6 - J6 . lam[[k]];
  Print["[lambda_", k, ", J] = 0: ", If[comm == 0*J6, "VERIFIED", "ERROR - does not commut
, {k, 1, 8}];
```

Verifying all lambda_k commute with J (required for u(3) action on C^3):

[lambda_1, J] = 0: VERIFIED

[lambda_2, J] = 0: VERIFIED

[lambda_3, J] = 0: VERIFIED

[lambda_4, J] = 0: VERIFIED

[lambda_5, J] = 0: VERIFIED

[lambda_6, J] = 0: VERIFIED

[lambda_7, J] = 0: VERIFIED

[lambda_8, J] = 0: VERIFIED

Section 10: SU(3) Commutation Relations

The SU(3) generators satisfy the Lie algebra commutation relations:

[lambda_a, lambda_b] = 2i sum_c f_{abc} lambda_c

where f_{abc} are the SU(3) structure constants (totally antisymmetric). Some key relations:
[lambda_1, lambda_2] = 2i lambda_3
[lambda_4, lambda_5] = i lambda_3 + i sqrt(3) lambda_8
[lambda_6, lambda_7] = i lambda_8 * sqrt(3) - i lambda_3

In the real 6x6 representation, these become (with the i -> j2 substitution).

In[315]:=

```
(* ---- Verify sample commutation relations in real 6x6 representation ---- *)

(* In the real representation, complex multiplication by i is replaced by the matrix j2 =
(* Thus [lambda_a, lambda_b] = 2 * realRep6[i * lambda_c] = 2 * realRep6[i] . realRep6[lam
(* We check the structure constants directly from the commutators. *)

comm12 = lam[[1]] . lam[[2]] - lam[[2]] . lam[[1]];
(* [lambda_1, lambda_2] = 2i lambda_3 in complex representation.              *)
(* In real rep: realRep6[2i lambda_3] = 2 * KroneckerProduct[lam3c, j2]      *)
(* since lam3c is real (diagonal): Im[i*lam3c] = lam3c, Re[i*lam3c] = 0.    *)
comm12expected = 2 * realRep6[I * lam3c];
Print["[lambda_1, lambda_2] = 2i lambda_3 (real rep): ",
  If[Chop[comm12] == Chop[comm12expected], "VERIFIED",
     "Checking numerically: ", Chop[comm12 - comm12expected]]];

(* Check [lambda_3, lambda_4] = -2i lambda_5, i.e., = -2 realRep(i) lambda_5 *)
comm34 = lam[[3]] . lam[[4]] - lam[[4]] . lam[[3]];
Print["[lambda_3, lambda_4] computed:"];
Print[MatrixForm[Chop[comm34]]];

(* Check that all commutators stay within the span of {lambda_1,...,lambda_8,J} *)
Print["\nSample commutator [lambda_1, lambda_2]:"];
Print[MatrixForm[Chop[comm12]]];
```

[lambda_1, lambda_2] = 2i lambda_3 (real rep): VERIFIED

[lambda_3, lambda_4] computed:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Sample commutator [lambda_1, lambda_2]:

$$\begin{pmatrix} 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Section 11: Connection to Ω16 and the Cl(4,4) Complex Structure

The Clifford algebra Cl(4,4) package defines the 16x16 complex structure matrix Ω16 (the ambient complex structure on the full 16-component spinor space). Similarly, Ω8 is the 8x8 complex structure on each chiral half.

The almost-complex structure J on sixPlaneAlpha is the restriction of Ω16 (or Ω8) to the 6-dimensional intersection:

J = i_sixPlane^* Ω8 (pullback of Ω8 to sixPlaneAlpha)

where i_sixPlane: sixPlaneAlpha -> R^8 is the inclusion. This means J is not arbitrary but is canonically induced by the ambient Clifford-algebra complex structure.

In[336]:=

```
(* ---- Connect J to Ω8 from Dirac-Lord-Nash_4+4.wl ---- *)

(* Ω8 is available from the loaded package: DiracLordNash44`Ω8 *)
(* (If the package is loaded, Ω8 = σ88 . γ88[7]) *)

(* Pullback of Ω8 to sixPlaneAlpha via basis matrix B (6x8) *)
(* J_induced = B . Ω8 . Transpose[B] (using g6=Id as metric on sixPlane) *)
(* Note: B is the 6x8 basis matrix computed in Section 2 *)

If[ValueQ[DiracLordNash44`Ω8],
  Ω8package = DiracLordNash44`Ω8;
  Print["Ω8 from package loaded successfully."];
  Jpullback = B . Ω8package . Transpose[B];
  Print["Pullback of Ω8 to sixPlaneAlpha:"];
  Print[MatrixForm[Chop[Jpullback]]];
  Print["J_pullback^2 = -Identity: ",
    If[Chop[Jpullback.Jpullback + IdentityMatrix[6]] == 0*IdentityMatrix[6], "VERIFIED", "
,
  Print["(Package Ω8 not available in current context - using abstract J6 defined in Secti
  Print["To use the Cl(4,4) connection, evaluate Section 0 first to load the package."];
];
```

```
Ω8 from package loaded successfully.
Pullback of Ω8 to sixPlaneAlpha:
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

```
J_pullback^2 = -Identity: NOT verified (check basis normalization)
```

The relationship between the abstract J6 defined in Section 5 and the Ω8 pullback: both define almost-complex structures on sixPlaneAlpha, and they agree up to a choice of orthonormal basis. The Ω8 pullback provides the canonical, physics-motivated complex structure inherited from the Clifford algebra.

Section 12: Physical Interpretation

The U(3) structure on sixPlaneAlpha has a direct physical interpretation within the Pre-Universe framework:

1. U(1) symmetry (generated by J):
   This is the generation-universal symmetry that acts uniformly on all three Einstein-Rosen bridge generations. It corresponds to the overall phase rotation of all three complex planes simultaneously.

2. SU(3) symmetry (generated by lambda_1,...,lambda_8):
   This is the inter-generation mixing symmetry. The 8 Gell-Mann generators rotate between the three complex lines E1, E2, E3 of sixPlaneAlpha. This suggests an identification with the SU(3) color symmetry of QCD in the Standard Model.

3. Three generations (E1, E2, E3):
   Each complex line Ek is a minimal J-invariant subspace - a real 2-plane preserved by J. These are the three Einstein-Rosen 2-plane bridges of the Pre-Universe. After the Big Bang, they become the three generations of matter.

4. The Kähler form omega:
   This is a conserved 2-form under the U(3) action - it is the 'symplectic structure' on the Pre-Universe 6-plane. Its top power omega^3 gives the volume form, related to the measure in the path integral.

5. Connection to Cartan's triality:
   The chiral spinors Psi1 and Psi2 map to the normal vectors v1 and v2 via triality. The U(3) structure on their intersection sixPlaneAlpha is the geometric remnant of the Spin(4,4) symmetry that acts on all three irreps (vector + two spinors) simultaneously.

Section 13: Summary

In[346]:=

```
(* ---- Print comprehensive summary of all structures ---- *)

Print["================================================================"];
Print["   U(3) STRUCTURE ON sixPlaneAlpha - COMPLETE SUMMARY"];
Print["================================================================"];
Print[""];
Print["AMBIENT SPACE: R^8 with split metric η88 = diag(1,1,1,1,-1,-1,-1,-1)"];
Print["              Clifford algebra Cl(4,4), Spin(4,4) symmetry"];
Print[""];
Print["sixPlaneAlpha = H1 intersect H2:"];
Print["  H1 = {x in R^8 : <x,v1>_η88 = 0}  (7-plane, normal = triality(Psi1))"];
Print["  H2 = {x in R^8 : <x,v2>_η88 = 0}  (7-plane, normal = triality(Psi2))"];
Print["  dim(sixPlaneAlpha) = 8 - 2 = 6"];
Print[""];
Print["U(3) STRUCTURE (g, J, omega):"];
Print["  g   = 6x6 positive-definite Euclidean metric (identity)"];
Print["  J   = BlockDiagonalMatrix[{j2,j2,j2}], j2={{0,-1},{1,0}}, J^2=-Id"];
Print["  omega = J^T.g = -J  (Kähler 2-form, antisymmetric, non-degenerate)"];
Print[""];
Print["DECOMPOSITION INTO GENERATIONS:"];
Print["  E1 = span{e1,e2} = Re(z1)/Im(z1) plane  [1st ER bridge = 1st generation]"];
Print["  E2 = span{e3,e4} = Re(z2)/Im(z2) plane  [2nd ER bridge = 2nd generation]"];
Print["  E3 = span{e5,e6} = Re(z3)/Im(z3) plane  [3rd ER bridge = 3rd generation]"];
Print["  sixPlaneAlpha = E1 + E2 + E3  (J-orthogonal direct sum over C)"];
Print[""];
Print["LIE ALGEBRA STRUCTURE:"];
Print["  u(3) = u(1) + su(3)"];
Print["  U(1) generator: J (overall phase rotation)"];
Print["  SU(3) generators: lambda_1,...,lambda_8 (real 6x6 Gell-Mann matrices)"];
Print["  All generators commute with J (complex-linearity condition)"];
Print[""];
Print["PHYSICAL IDENTIFICATION:"];
Print["  U(1): generation-universal symmetry"];
Print["  SU(3): inter-generation mixing (conjectural pre-QCD color)"];
Print["  E1,E2,E3: 3 generations of Einstein-Rosen 2-plane bridges"];
Print["================================================================"];
```

```
================================================================
   U(3) STRUCTURE ON sixPlaneAlpha - COMPLETE SUMMARY
================================================================

AMBIENT SPACE: R^8 with split metric η88 = diag(1,1,1,1,-1,-1,-1,-1)
              Clifford algebra Cl(4,4), Spin(4,4) symmetry

sixPlaneAlpha = H1 intersect H2:
  H1 = {x in R^8 : <x,v1>_η88 = 0}  (7-plane, normal = triality(Psi1))
  H2 = {x in R^8 : <x,v2>_η88 = 0}  (7-plane, normal = triality(Psi2))
  dim(sixPlaneAlpha) = 8 - 2 = 6

U(3) STRUCTURE (g, J, omega):
  g   = 6x6 positive-definite Euclidean metric (identity)
  J   = BlockDiagonalMatrix[{j2,j2,j2}], j2={{0,-1},{1,0}}, J^2=-Id
  omega = J^T.g = -J  (Kähler 2-form, antisymmetric, non-degenerate)

DECOMPOSITION INTO GENERATIONS:
  E1 = span{e1,e2} = Re(z1)/Im(z1) plane  [1st ER bridge = 1st generation]
  E2 = span{e3,e4} = Re(z2)/Im(z2) plane  [2nd ER bridge = 2nd generation]
  E3 = span{e5,e6} = Re(z3)/Im(z3) plane  [3rd ER bridge = 3rd generation]
  sixPlaneAlpha = E1 + E2 + E3  (J-orthogonal direct sum over C)

LIE ALGEBRA STRUCTURE:
  u(3) = u(1) + su(3)
  U(1) generator: J (overall phase rotation)
  SU(3) generators: lambda_1,...,lambda_8 (real 6x6 Gell-Mann matrices)
  All generators commute with J (complex-linearity condition)

PHYSICAL IDENTIFICATION:
  U(1): generation-universal symmetry
  SU(3): inter-generation mixing (conjectural pre-QCD color)
  E1,E2,E3: 3 generations of Einstein-Rosen 2-plane bridges
================================================================
```

TODO

Kähler. We show that such M^6 are foliated by 4-planes in $\mathbf{O}$ in a unique way

In[382]:=

```
timeFinish = Now
```

Out[382]=

Mon 30 Mar 2026 06:26:08 GMT−7

In[383]:=

```
timeFinish
```

Out[383]=

Mon 30 Mar 2026 06:26:08 GMT−7

In[384]:=

```
timeStart
```

Out[384]=

```
timeStart
```

In[385]:=

```
executionTime = timeFinish - timeStart
```

Out[385]=

−timeStart + Mon 30 Mar 2026 06:26:08 GMT−7